

Bill of Materials - Electrical					
Value	Part Number	DigiKey Number	Manufacturer	Designator	Quantity
CAP CER 15000 pF	885342208013	732-12124-1-ND	Würth Elektronik	C1, C4	1
CAP CER 22 uF	GRM32ER71E226ME15L	490-10758-1-ND	Murata	C2	2
CAP CER 10 uF	GRM32ER71H106MA12L (CL32B106KBJNNNE)	1276-3387-1-ND	Samsung	C3	1
CAP CER 2.2 uF	GCM31CL8EL225KE07L	490-GCM31CL8EL225KE07LCT-ND	Murata	C5, C6	2
CAP CER 0.1 uF	GCM31C5C1H104FA16L	490-GCM31C5C1H104FA16LCT-ND	Murata	C7, C8, C11, C19, C21	5
CAP CER 10000 pF	GRM31B7U2E103JW31L	490-6502-1-ND	Murata	C9	1
CAP CER 4.7 uF	GRM21BR71A475KE51L	490-GRM21BR71A475KE51LCT-ND	Murata	C10	1
CAP CER 0.1 uF	VJ1206Y104KXXAC	720-VJ1206Y104KXXACCT-ND	Vishay Vitrumon	C12, C13, C14, C15, C16	5
CAP CER 15 pF	GRM31A5C2H150JW01D	490-14667-1-ND	Murata	C17, C18	2
CAP CER 0.1 F	C1206X104G4HAC7800 (C1206X104G4HACTU)	399-C1206X104G4HACTUCT-ND	KEMET	C20, C22, C23, C24, C26, C27, C29, C31, C34, C35, C37, C39, C40, C41, C42, C43, C44, C45, C46, C47, C48, C49	22
CAP CER 10 uF	GCM31CM8EG106KE08L	490-GCM31CM8EG106KE08LCT-ND	Murata	C25, C28, C30, C32, C33, C36	6
CAP CER 2.2 uF	CC1206ZRY5V8BB225	311-3464-1-ND	Yageo	C38	1
TVS Diode 5.5 V	TPD1E10B06DPYR	296-30406-1-ND	TI	D1, D2	2
TVS SOID123FL 5 V	SMF5.0A	F5748CT-ND	Littlefuse	D3, D4	2
TVS Diode 5.25 V	USBLC6-4SC6	497-4492-1-ND	STMicroelectronics	D5	1
Green LED, 20mA	304090040	N/A	Seeed Studio	DS1	1
PTC Reset Fuse 13.2 V	MICROSMD050F-2	MICROSMD050F-2CT-ND	Littlefuse	F1	1
5V-5V DC Isolator	UCC33420QRAQRQ1	296-UCC33420QRAQRQ1TR-ND	TI	IC1	1
USB-C 2.0 CON	USB4105GFA060	2073-USB4105-GF-A-060CT-ND	GCT	J1	1
5-pin CON Header	S5B-XH-A-1(LF)(SN)	455-4228-ND	JST	J2	1
20(2x10) ZIF Socket	220-3342-00-0602J	3M2002-ND	3M	J3	1
RES 1 K ohm 1%	RC1206FR-101KL	13-RC1206FR-101KLCT-ND	Yageo	R1	1
RES 5.1 K ohm 0.1%	RP1206BRD075K1L	13-RP1206BRD075K1LCT-ND	Yageo	R2, R3	2
RES 920 ohm	RN73R1JTDD9200B25	2019-RN73R1JTDD9200B25CT-ND	KOA Spear	R4	1
RES 1.69 K ohm 0.1%	RQ73C2B1K69BTD	2019-RN73R2ATDD5561B25CT-ND	TE Connectivity	R5	1
RES 442 ohm 0.1%	RT1206BRD07442RL	13-RT1206BRD07442RLCT-ND	Yageo	R6	1
RES 5.56 K ohm 0.1%	RN73R2ATDD5561B25	2019-RN73R2ATDD5561B25CT-ND	KOA Spear	R7	1
RES 2.49 K ohm 1%	KTR18E2PF2491 (RMCF1206FT2K49CT)	RMCF1206FT2K49CT-ND	ROHM	R8, R12, R13, R14, R15, R16, R18	7
RES Thick Film 57.6 K ohm 1%	CRCW120657K6FKEA	541-57.6KFCT-ND	Vishay Dale	R9	1
RES 4.17 K ohm 0.1%	RN73R1JTDD4171B25 (RN73H1JTDD4171F100)	2019-RN73H1JTDD4171F100CT-ND	KOA Spear	R10	1
RES 2.55 K ohm 0.1%	RT1206BRD072K55L	YAG5059CT-ND	Yageo	R11	1
RES 1.13 K ohm 1%	RK73H2BTDD1131F	2019-RK73H2BTDD1131FCT-ND		R17	1
RES 5 K ohm 0.1%	RT1206BRD075KL	YAG5090CT-ND	Yageo	R19	1
RES 10 K ohm 0.1%	RP1206BRD0710KL	13-RP1206BRD0710KLCT-ND	Yageo	R20, R23	2
RES 2.2 K ohm 1%	RK73H2ETDD2201F	2019-RK73H2ETDD2201FCT-ND	KOA Spear	R21, R22, R24, R25	4
RES 125 ohm 0.1%	RT1206BRD07125RL	13-RT1206BRD07125RLCT-ND	Yageo	R26	1
RES 10 K ohm 0.1%	TNPW120610K0BET	541-3924-1-ND	Vishay Dale	R27, R28, R30	3
RES 20 K ohm 0.1%	TNPW120620K0BYEN	541-4422-1-ND	Vishay Dale	R29	1
RES 4.7 K ohm 0.1%	TNPW12064K70BEEA	541-3881-1-ND	Vishay Dale	R31, R32	2

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RES 100 K ohm 0.1%	TNPW1206100KBEEA	TNP100KACCT-ND	Vishay Dale	R33, R34	2
USB Digital Isolator	ADUM4160BRWZ	505-ADUM4160BRWZ-ND	Analog Devices	U1	1
IC REG Linear 3.3 V	TPS7A2033PDBVR	296-TPS7A2033PDBVRCT-ND	TI	U2	1
CP2102 TH, female USB-C	CP2102-GM	336-1160-5-ND	Silicon Labs	U3	1
TLV431 (1.24 V reference)	TLV431ALPG	TLV431ALPGOS-ND	onsemi	U4, U5, U6, U7, U8	5
64-pin 48 MHz	STM32C071RBT6	497-STM32C071RBT6-ND	STMicroelectronics	U9	1
8:1 Multiplexer	ADG708BRUZ-REEL	ADG708BRUZ-REELCT-ND	Analog Devices	U10, U15, U18, U20	4
IC CROSSPOINT SWITCH 8X12 32LFCSP	ADG2128BCPZ-REEL7	505-ADG2128BCPZ-REEL7CT-ND	Analog Devices	U12, U13	2
2W ISO DC-DC Converter	DCP020509P	DCP020509P-ND	TI	U14	1
4:1 Multiplexer	ADG709BRUZ-REEL	ADG709BRUZ-REELCT-ND	Analog Devices	U16	1
IC ADC 24bit Sigma-Delta 16DIP	NAU7802KGI-ND	NAU7802KGI-ND	Nuvoton	U17	1
24 MHz XTAL 10-1-U-T	ABM3B-24.000MHZ-10-1-U-T	300-8216-1-ND	STMicroelectronics	Y1	1
ST-LINK STM Flasher	ST-LINK/V2	511-ST-LINK/V2	Chip Quick	N/A	1

Bill of Materials - Chassis

Name	Part #	Manufacturer Number	Manufacturer	Designator	Quantity
ABS Enclosure 2-Panels	JB-65T*0000	JB-65T*0000	Polycase	1, 2, 3, 4, 5	1
Hex Standoff #4-40 1"	101604HFA	4753-101604HFA-ND	Kanebridge	6	4
Mach Screw Pan #4-40	R4-40X1/4 2701	335-1082-ND	APM Hexseal	7	8
Bumper Cylindrical 0.5"	SJ-5012 (BLACK)	SJ5012-0-ND	3M	8	4